

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 Cyber & Information Security Practical

PRACTICAL - 3

Aim :- Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity. **(CLI)**

Code :-

```
# T101 Practical 3
# Message Authentication Code (MAC) using HMAC-SHA256
# CLI Version
import hmac
import hashlib

def generate_mac(message, key):
    mac = hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()
    return mac

print("=" * 50)
print("  MESSAGE AUTHENTICATION CODE (MAC)")
print("=" * 50)
message = input("Enter Message : ")
key = input("Enter Secret Key : ")
mac = generate_mac(message, key)
print("\nGenerated MAC:")
print(mac)
print("Verification")
received_message = input("Enter Received Message : ")
received_key = input("Enter Secret Key : ")
received_mac = input("Enter Received MAC : ")
new_mac = generate_mac(received_message, received_key)
if hmac.compare_digest(new_mac, received_mac):
    print("\nMessage Verified Successfully")
    print("Integrity Maintained")
    print("Sender is Authentic")
else:
    print("\nVerification Failed")
    print("Message has been Modified or Key is Incorrect")
```

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Output :-

```
*IDLE Shell 3.14.6*
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-52/Documents/T101 Cyber Practical/T101 Practical - 3 (CLI).py =====
=====
MESSAGE AUTHENTICATION CODE (MAC)
=====
Enter Message : T101 RAJDEEP M PARAB
Enter Secret Key : 101

Generated MAC:
35f1b92743e8d32be556d52e60081c6010d9da9336e84ded1cf5ebc508e0c31e
Verification
Enter Received Message : T101 RAJDEEP M PARAB
Enter Secret Key : 101
Enter Received MAC : 35f1b92743e8d32be556d52e60081c6010d9da9336e84ded1cf5ebc508e0c31e
...
Message Verified Successfully
Integrity Maintained
Sender is Authentic
>>>
===== RESTART: C:/Users/IT-52/Documents/T101 Cyber Practical/T101 Practical - 3 (GUI).py =====
>>>
===== RESTART: C:/Users/IT-52/Documents/T101 Cyber Practical/T101 Practical - 3 (GUI).py =====
|
```

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Aim :- Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity. **(GUI)**

Code :-

```
# T101 Practical 3
# Message Authentication Code (MAC) using HMAC-SHA256
# GUI Version
import tkinter as tk
from tkinter import messagebox
import hashlib
import hmac

def generate_mac():
    message = entry_message.get()
    key = entry_key.get()
    if message == "" or key == "":
        messagebox.showerror("Error", "Enter Message and Secret Key")
        return
    mac = hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()
    entry_mac.delete(0, tk.END)
    entry_mac.insert(0, mac)

def verify_mac():
    message = entry_verify_message.get()
    key = entry_verify_key.get()
    received_mac = entry_verify_mac.get()
    new_mac = hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()
    if hmac.compare_digest(new_mac, received_mac):
        lbl_result.config(
            text="Message Verified Successfully",
            fg="green"
        )
    else:
        lbl_result.config(
            text="Verification Failed",
            fg="red"
        )

root = tk.Tk()
root.title("Message Authentication Code (MAC)")
root.geometry("750x600")
root.configure(bg="#eef6ff")
title = tk.Label(
    root,
    text="Message Authentication Code (HMAC-SHA256)",
    font=("Arial", 18, "bold"),
    bg="#eef6ff",
```

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```
fg="navy"
)
title.pack(pady=15)
frame1 = tk.LabelFrame(
    root,
    text="Generate MAC",
    padx=15,
    pady=15,
    bg="#eef6ff"
)
frame1.pack(fill="x", padx=20)
tk.Label(frame1, text="Message", bg="#eef6ff").grid(row=0, column=0, pady=8)
entry_message = tk.Entry(frame1, width=55)
entry_message.grid(row=0, column=1)
tk.Label(frame1, text="Secret Key", bg="#eef6ff").grid(row=1, column=0, pady=8)
entry_key = tk.Entry(frame1, width=55, show="*")
entry_key.grid(row=1, column=1)
tk.Button(
    frame1,
    text="Generate MAC",
    command=generate_mac,
    bg="green",
    fg="white",
    width=18
).grid(row=2, column=1, pady=10)
tk.Label(frame1, text="Generated MAC", bg="#eef6ff").grid(row=3, column=0)
entry_mac = tk.Entry(frame1, width=70)
entry_mac.grid(row=3, column=1)
frame2 = tk.LabelFrame(
    root,
    text="Verify MAC",
    padx=15,
    pady=15,
    bg="#eef6ff"
)
frame2.pack(fill="x", padx=20, pady=15)
tk.Label(frame2, text="Message", bg="#eef6ff").grid(row=0, column=0, pady=8)
entry_verify_message = tk.Entry(frame2, width=55)
entry_verify_message.grid(row=0, column=1)
tk.Label(frame2, text="Secret Key", bg="#eef6ff").grid(row=1, column=0)
entry_verify_key = tk.Entry(frame2, width=55, show="*")
entry_verify_key.grid(row=1, column=1)
tk.Label(frame2, text="Received MAC", bg="#eef6ff").grid(row=2, column=0)
entry_verify_mac = tk.Entry(frame2, width=70)
entry_verify_mac.grid(row=2, column=1)
tk.Button(
    frame2,
    text="Verify MAC",
    command=verify_mac,
    bg="blue",
    fg="white",
    width=18
).grid(row=3, column=1, pady=10)
```

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```
lbl_result = tk.Label(  
    root,  
    text="",  
    font=("Arial", 14, "bold"),  
    bg="#eef6ff"  
)  
lbl_result.pack(pady=20)  
root.mainloop()
```

Output :-

